



BUILDING MANAGEMENT SYSTEM PROPOSAL

CSIR Campus Building 53 Head Office Upgrade Project

Proposed BMS Architecture, HVAC Control Strategy & Implementation
Approach

PREPARED FOR
CSIR

COMPANY
Spartcon

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DATE
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Department of Science & Innovation, South Africa · Basis of design derived from Building 53 HVAC mechanical drawings (issued for construction 07-05-2026).

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How to read this proposal. Throughout this document, technical statements are qualified against their level of certainty so that confirmed information is never presented as assumption, and engineering proposals are never presented as fact.

- CONFIRMED Established directly from the Building 53 HVAC drawings.
- OEM Supported by manufacturer capability, subject to model selection.
- ENGINEERING Proposed engineering basis of design.
- CONFIRM Requires client / OEM / HVAC contractor confirmation.

This proposal is based on the HVAC mechanical drawing set for Building 53. It does not constitute a complete BMS controls specification; the drawings do not define BMS network, controller I/O, sensor schedules or control sequences. Those items are addressed here as a proposed basis of design and are identified as requiring confirmation during detailed design.

01

Executive Summary & Project Understanding

Spartcon has been asked to propose a Building Management System (BMS) for the CSIR Campus Building 53 – Head Office Upgrade Project. The proposed BMS will provide centralised monitoring and control of selected HVAC equipment through a network of field controllers and standard building-automation communication protocols, presented to operators through a web-based interface.

This proposal is developed from the HVAC mechanical drawing set for Building 53, comprising the Ground, First, Second and Third Floor HVAC plans, the Roof Plan HVAC, and associated plot/log information. That set was issued for construction on 07 May 2026.

What the drawings confirm

The drawings are primarily mechanical HVAC documents. They confirm the presence of the following HVAC infrastructure and equipment:

- GMS ducting
 - VAV diffusers
 - Return air grilles
 - Ablution extract / disc valves
- High-wall split air-conditioning units
 - Outdoor heat pump / air-conditioning units
 - Refrigerant piping
 - Condensate / drain arrangements and HVAC construction details

What the drawings do not define

The drawing set does not contain a complete BMS controls specification. There is no confirmed information for the BMS network (BACnet or Modbus), controller platform, controller I/O schedules, sensor schedule, control sequences, actuator communication protocols, lighting-control network, network topology, remote-access architecture or final BMS point count. These items are treated in this proposal as a proposed engineering basis of design and are clearly identified as requiring confirmation.

Basis of the proposal. The HVAC consultant/designer identified on the mechanical drawings is not the author of this BMS proposal and is not associated with Spartcon. This document is prepared by Spartcon as an independent BMS proposal.

Proposed direction

The proposed solution is based around a **Vector Controls** architecture, treated as its own controls ecosystem rather than a Tridium JACE / Niagara system. Field devices connect to Vector X2 field controllers over BACnet MS/TP or Modbus RTU, which connect over the building Ethernet network to a Vector management platform serving a web-based operator interface. Controller models, quantities and the final management platform are established from the approved I/O schedule during detailed design.

PROJECT PARAMETER	POSITION IN THIS PROPOSAL
Project	CSIR Campus Building 53 – Head Office Upgrade Project CONFIRMED
Source information	Building 53 HVAC mechanical drawings, issued for construction 07-05-2026 CONFIRMED
Controls ecosystem	Vector Controls (Vector X2 field controllers) ENGINEERING
Management platform	Vector BAS-SMP-2000 as proposed basis; GSM-1000-BMX noted as alternative CONFIRM
BMS point count / I/O	To be established from the approved equipment schedule CONFIRM

02

Proposed BMS Solution

The proposed BMS is a Vector-based control architecture. Field devices and HVAC equipment interfaces are connected to Vector X2 field controllers, which communicate over standard building-automation buses to a Vector management platform. Operators interact with the system through a browser-based interface for monitoring, alarms, trends and scheduling.

Vector X2 field controllers

Vector X2 controllers are presented as programmable communicating field controllers within the Vector controls ecosystem. Depending on the controller model selected, relevant communication capabilities include:

- BACnet MS/TP
- BACnet/IP
- Modbus RTU / ASCII
- Modbus TCP

Controller configuration and programming are associated with the Vector EasySet environment. Controller model selection and quantities are not fixed in this proposal; they will be determined from the final I/O schedule and the equipment interfaces confirmed during detailed design.

Management platform

The proposed basis of design for the management layer is the **Vector BAS-SMP-2000**. A second Vector management product, **GSM-1000-BMX**, is noted as a possible alternative subject to final system selection. No single management platform is treated as finally selected; the platform above is stated as the proposed basis only.

Vector compared with a Tridium JACE / Niagara system

This distinction is important and is stated precisely:

TRIDIUM JACE / NIAGARA	VECTOR CONTROLS
A JACE is a hardware/network automation controller that runs the Niagara framework.	Vector X2 controllers are independent field controllers belonging to the Vector controls ecosystem.
Niagara Workbench is used to engineer and configure Niagara systems.	Vector has its own configuration environment (Vector EasySet). Vector is not used to program a JACE.
Communicates using standard protocols including BACnet and Modbus.	Communicates using standard protocols including BACnet and Modbus, and can therefore operate as an independent BMS/control architecture.

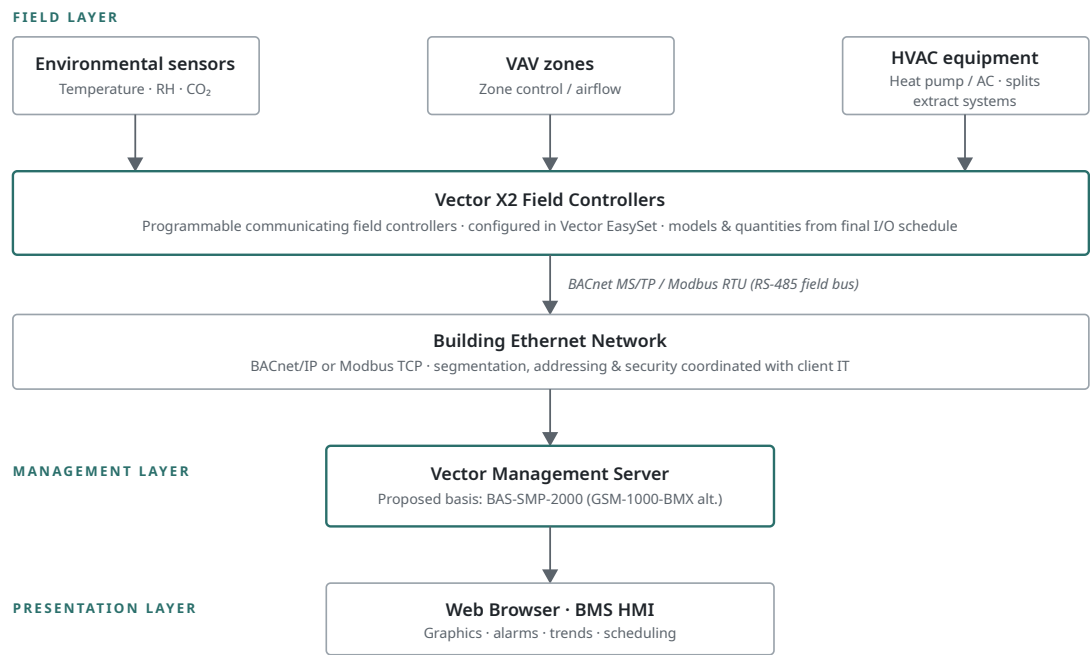
Because both platforms use standard communication protocols, a Vector + Niagara/JACE hybrid architecture is technically possible where appropriate. However, this proposal is primarily evaluating a Vector-based architecture. No official partnership between Vector and Tridium is implied or claimed.

Scope note. The described capabilities reflect the Vector product ecosystem. Final controller models, the management platform, and the exact protocol used per equipment item remain subject to confirmation during detailed design and against the approved equipment schedule.

03

System Architecture

The proposed architecture is a conventional layered building-automation topology: field devices report to field controllers, field controllers communicate over field buses to the management layer, and the management layer presents a web-based interface. The diagram below shows the proposed conceptual architecture.



Vector-supplied elements (proposed basis) Field / third-party / client-provided elements

Diagram is conceptual. The final physical network architecture, device addressing and protocol assignment per equipment item are coordinated with the client IT/network team and confirmed against the approved equipment schedule during detailed design.

04

HVAC Control Strategy

The following proposed control strategy describes the intended BMS functions per equipment type. In every case the final control points depend on the selected equipment, its controller/actuator and its available communication interface, and are confirmed during detailed design.

VAV systems

Where the installed VAV equipment supports suitable control interfaces, the BMS should be capable of:

- Monitoring zone temperature
- Monitoring relevant zone conditions
- Adjusting temperature/setpoint parameters where supported
- Monitoring VAV operating status
- Monitoring airflow-related values where available
- Managing occupied/unoccupied operation where applicable
- Generating alarms for abnormal conditions
- Trending relevant values

No particular actuator or communication protocol is assumed for any VAV. Final control points depend on the selected VAV controller/actuator. **CONFIRM**

Temperature / humidity / CO₂

The proposed system supports environmental sensors for temperature, relative humidity and CO₂. Where required by the final design, these values can be used for monitoring, alarming, trending and setpoint/control strategies. Exact sensor quantities and acceptable limits are not defined at this stage.

CONFIRM

Heat pump / air-conditioning systems

The drawings identify outdoor heat pump / AC equipment with the following ratings taken from the mechanical drawings:

EQUIPMENT	RATING (FROM DRAWINGS)	STATUS
HP/AC01	302.1 kW	CONFIRMED (rating from drawings)
HP/AC02	585.8 kW	CONFIRMED (rating from drawings)

Manufacturer, model, protocol and exact control interface are not defined by the drawings. BMS integration will depend on the equipment OEM interface and may use BACnet, Modbus, dry contacts, gateway integration or another approved interface as applicable. Potential monitoring/control functions — all subject to OEM confirmation — may include enable/disable where supported, run status, fault/alarm status, operating mode, temperature values, setpoints and relevant equipment parameters. **OEM**

04

HVAC Control Strategy (continued)

Split air-conditioning units

The drawings include high-wall split units. Integration depends on the selected split-unit manufacturer's available communication interface. Possible approaches include:

- Direct BACnet / Modbus interface
- Manufacturer gateway
- Approved third-party interface
- Monitoring-only integration where full control is unavailable

Direct integration is not claimed without confirmation of the selected manufacturer's interface. **OEM**

Extract systems

The proposal addresses monitoring/control of relevant extract systems where applicable. Potential points include:

- Run status
- Fault status
- Enable/disable
- Scheduling

Final points depend on the installed equipment. **CONFIRM**

Summary of proposed points by equipment type

EQUIPMENT	TYPICAL PROPOSED FUNCTIONS	DEPENDENCY
VAV zones	Zone temperature, zone conditions, setpoint adjust (where supported), status, airflow (where available), occupancy mode, alarms, trends	Selected VAV controller/actuator
Environmental sensors	Temperature, RH, CO ₂ — monitoring, alarming, trending, control strategies	Final design / sensor selection
Heat pump / AC (HP/AC01, HP/AC02)	Enable/disable, run status, fault, mode, temperatures, setpoints, parameters	OEM interface
Split AC units	Monitoring and control per available interface; monitoring-only where control unavailable	Manufacturer interface / gateway
Extract systems	Run status, fault, enable/disable, scheduling	Installed equipment

Point allocation. The functions above are a proposed control strategy. Final point allocation will be established from the approved equipment schedule and confirmed control philosophy during detailed design.

05

Monitoring, Alarms, Trends & Scheduling

Graphics

A browser-based graphical interface is provided for building overview, floor-level views, HVAC equipment, VAV zones, environmental conditions, equipment status and alarms. Graphics reflect the equipment confirmed in the drawings and the points established during detailed design.

Alarms

Centralised alarms are provided for conditions such as equipment faults, communication failures, sensor faults, abnormal operating conditions and other relevant HVAC alarms. Alarm points and thresholds are confirmed against the final control philosophy.

Trends

Historical trending is provided for relevant temperature, humidity, CO₂, equipment status, setpoints and other selected BMS points.

Scheduling

Schedules may be used for occupied/unoccupied operation, HVAC enable/disable, equipment operating periods and other approved schedules.

Data / reporting

Where supported by the selected management platform, the system provides historical data, alarm history, trends, operating information and basic reporting. Advanced analytics are not promised and are only included where specifically supported by the selected platform and agreed in scope.

FUNCTION	DESCRIPTION	BASIS
Graphics	Building overview, floor views, HVAC equipment, VAV zones, conditions, status, alarms	ENGINEERING
Alarms	Equipment faults, comms failures, sensor faults, abnormal conditions, HVAC alarms	ENGINEERING
Trends	Temperature, humidity, CO ₂ , status, setpoints, selected points	ENGINEERING
Scheduling	Occupied/unoccupied, HVAC enable/disable, operating periods	ENGINEERING
Data / reporting	Historical data, alarm history, trends, operating info, basic reporting	Platform-dependent

06

Communications & Network Architecture

Field controllers communicate with field devices over RS-485 field buses using BACnet MS/TP or Modbus RTU. The field controllers connect to the building Ethernet network, over which they communicate with the Vector management server using BACnet/IP or Modbus TCP. The management server presents the web-based operator interface.

The final physical network architecture must be coordinated with the client's IT/network team. The following considerations apply and are confirmed during detailed design:

CONSIDERATION	APPROACH
IP addressing	Allocated in coordination with client IT. No IP addresses are assigned in this proposal.
Network segmentation / VLANs	Applied where required, coordinated with client IT policy.
Switch infrastructure	Confirmed against available building network infrastructure and BMS requirements.
RS-485 field buses	Field bus layout, termination and biasing per device count and cable runs.
Device addressing	BACnet/Modbus device addressing scheme defined in the I/O schedule.
Network commissioning	Bus and IP communication verified during commissioning.
Cybersecurity	Aligned with client requirements. Controllers are not exposed directly to the public internet.
Remote access	Provided only through an approved, secured method per client requirements.

Security position. Controllers will not be exposed directly to the public internet. Remote access, segmentation and cybersecurity measures are implemented per the client's IT and security requirements, which are to be confirmed.

Existing building network availability, IT/VLAN requirements, remote-access requirements and client cybersecurity requirements are all subject to confirmation. **CONFIRM**

07 Implementation & Commissioning

Implementation follows a staged methodology. Each stage produces defined outputs that feed the next, so that hardware and programming decisions are grounded in a confirmed I/O schedule and control philosophy rather than assumptions.

STAGE	ACTIVITY	SCOPE
1	Detailed design	Review final HVAC equipment schedule; confirm equipment manufacturers/models; confirm communication interfaces; develop I/O schedule; confirm network requirements; confirm control philosophy.
2	Hardware / panel preparation	Controller selection; sensor selection; actuator/interface selection; panel and network requirements.
3	Installation	Field controllers; sensors; communication cabling; network connections; equipment interfaces.
4	Programming	Controller configuration; control sequences; alarms; scheduling; graphics; trends.
5	Commissioning	Point-to-point testing; sensor verification; equipment command verification; alarm testing; communication testing; functional testing.
6	Handover	As-built information; controller/database backup; configuration documentation; operator orientation/training; final commissioning records.

Dependency. Stage 1 outputs — the approved equipment schedule, confirmed interfaces, I/O schedule and control philosophy — determine controller models, quantities, sensor counts and final point allocation. These are not fixed in this proposal.

Programme durations, resourcing and sequencing are confirmed on award and against the project construction programme.

Assumptions, Exclusions & Information Required

This proposal is a basis of design developed from HVAC mechanical drawings. The following items are not confirmed by the drawings and require confirmation before detailed implementation. No prices, equipment quantities, compliance requirements or client standards are fabricated in this document.

Requires confirmation

- Final BMS point count
- Exact Vector controller models
- Exact controller quantities
- Sensor quantities
- VAV controller/actuator interfaces
- HVAC OEM communication protocols
- Split-unit integration
- Heat pump / AC integration
- Existing building network availability
- IT / VLAN requirements
- Remote access requirements
- Lighting-control requirements
- Client cybersecurity requirements
- Final BMS server / management platform selection
- Electrical power requirements
- BMS panel requirements & final control sequences

Assumptions

- The HVAC equipment identified in the Building 53 drawings represents the equipment to be integrated, subject to the final approved equipment schedule.
- Selected HVAC equipment provides accessible communication interfaces or points suitable for BMS integration; interface type is confirmed per OEM.
- A suitable building network is made available, or its requirements are agreed with client IT.

Exclusions

- Mechanical HVAC supply, installation and associated works.
- OEM equipment, gateways or licences not confirmed in the agreed BMS scope.
- Building IT network infrastructure beyond the BMS interface, unless specifically agreed.
- Electrical power provision to BMS panels and equipment, unless specifically agreed.
- Lighting-control scope, unless specifically agreed and defined.

Information required from client / consultant / HVAC contractor

Approved HVAC equipment schedule with manufacturers and models; OEM communication interface details for heat pump/AC, split units and extract systems; VAV controller/actuator details; building network and IT/security requirements; remote-access requirements; and confirmation of any lighting-control scope.

09

Commercial Information & Sign-off

PRELIMINARY — TO BE COMPLETED

The table below is a preliminary commercial structure to be completed once the equipment schedule, I/O schedule and control philosophy are confirmed. No prices, quantities, labour rates or costs are stated at this stage.

ITEM	DESCRIPTION	QTY	UNIT RATE	AMOUNT
BMS controllers	To be confirmed	TBC	TBC	TBC
Sensors	To be confirmed	TBC	TBC	TBC
BMS management system	Proposed Vector platform	TBC	TBC	TBC
Installation	Field installation	TBC	TBC	TBC
Programming & commissioning	BMS configuration and commissioning	TBC	TBC	TBC

This table is a placeholder for commercial information to be completed later. It does not represent a quotation.

Conclusion

This proposal sets out a Vector-based BMS approach for CSIR Campus Building 53, developed from the available HVAC mechanical drawings. It describes the proposed architecture, HVAC control strategy, BMS functions, network approach and staged implementation, while clearly separating what is confirmed by the drawings from what remains a proposed engineering basis of design. The information identified as requiring confirmation should be resolved with the client, OEMs and HVAC contractor during detailed design, after which controller models, quantities, point counts and commercial values can be finalised.

Acceptance & Sign-off

FOR AND ON BEHALF OF SPARTCON

FOR AND ON BEHALF OF CSIR

NAME

NAME

SIGNATURE

SIGNATURE

DATE

DATE

Prepared by Tshepo Mamabolo, Spartcon · Prepared for CSIR · 18 September 2026 · CSIR Campus Building 53 – Head Office Upgrade Project.